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# Fatigue life analysis of double-row tapered roller bearing in a modern wind turbine under oscillating external load and speed

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## Abstract

Fatigue life analysis of roller bearing is usually performed for bearings under constant rotating speed and invariant loading conditions. For the bearings used in offshore floating direct-drive wind turbines, they often experience oscillating motions with varying loading patterns, for which the standard fatigue life analysis is not valid due to the presence of fluctuating loads. This paper presents the fatigue life analysis of a double-row tapered roller bearing under oscillating external load and speed conditions, which is used to support the main shaft of a large modern direct-drive wind turbine. First, a comprehensive quasi-static model of the double-row tapered roller bearing is developed for determining the internal load distribution of rollers. The contact pressure of rollers is then studied using an iterative scheme based on the elastic contact model. After that, the formulation of basic rating life of the double-row tapered roller bearing with oscillating external load and speed is given to calculate the fatigue life. Numerical simulations are carried out to investigate the effects of the oscillating load and speed, angular misalignment, and internal clearance on the fatigue life of the bearing.

## Keywords

Fatigue life, double row tapered roller bearing, offshore wind turbine, clearance, angular misalignment, oscillating load, rotating speed

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## Introduction

Wind energy industry has been growing rapidly due to the cost-effective and environmentally friendly nature of wind energy among various energy resources.[1] In a large modern floating direct-drive wind turbine, a double-row tapered roller bearing (DTRB) and a cylindrical roller bearing (CRB) are commonly equipped in the front and rear of the main shaft as shown in Figure 1.[2] These bearings provide the support for the whole weight of the rotor assembly including the generator, main shaft, nacelle, and blades. Hence, theses supporting bearings are one of the key components and their fatigue failure is a critical issue dominating the operation of wind turbines.[3] It is noted that the failure rate of the main front DTRB is commonly higher than that of the rear CRB, because the front DTRB needs to support major radial and thrust loads.[4] Therefore, the prediction of the fatigue failure of the front DTRB of wind turbines is of particular practical importance to ensure a safe operation. An accurate estimation of the fatigue life of the main DTRB can provide useful information for predictive maintenance, which can greatly reduce the unscheduled and unpredicted downtime costs of wind turbines.[5]

Theoretical analysis of the fatigue life of rolling element bearings was originated from 1950s, pioneered by Lundberg and Palmgren (LP).[6–16] The theory of LP has now become a basis for modern calculation of fatigue life including ISO standards.[7–9] Since then, significant efforts have been made by researchers in estimating the fatigue life of roller bearings.[10–12] For tapered roller bearings (TRBs), Zantopulus[13] conducted an experimental study of the fatigue life of TRB by considering angular

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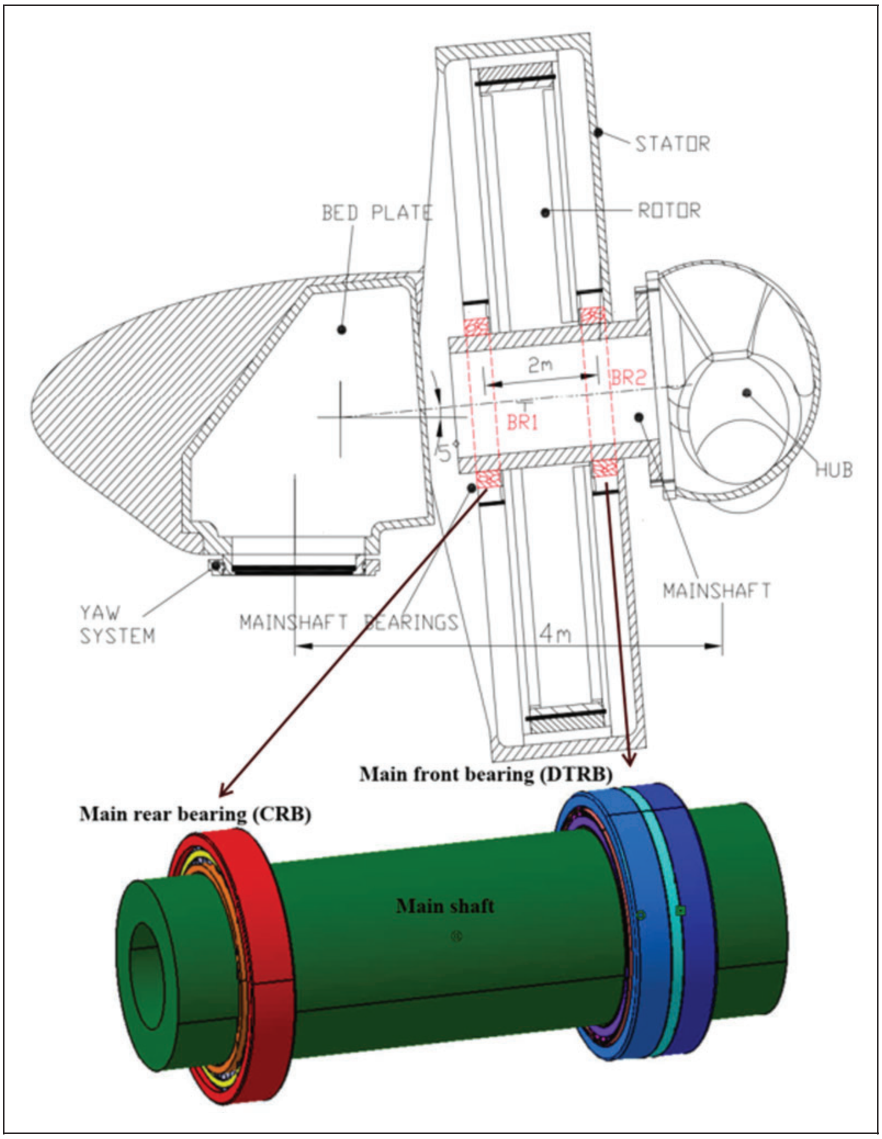


**Figure 1.** Schematic of the integrated mechanical drive-train system in a large modern direct-drive wind turbine, and a 3D view of the main shaft which is supported by a DTRB in the front and a CRB in the rear of the shaft.[2]

misalignment effect. Andreason[14] analysed internal load distribution and rating life of a TRB by taking misalignment angle, initial preload, and clearance into account. Liu[15] studied the load distribution and the fatigue life of a TRB under high-speed and combined external loads. Kabus et al.[16] estimated the basic reference life of a TRB using a six degrees-of-freedom (DOF) model. Tong and Hong[17,18] numerically investigated the effects of angular misalignment and external load on the basic rating life of a TRB by using a five degrees-of-freedom (DOF) bearing model, which was extended from the model proposed by de Mul et al.[19] Kotzalas[20] carried out an experimental study on the statistical distribution of TRB fatigue life. Gurumoorthy and Ghosh[21] experimentally investigated the failure of TRB used in the front wheel of a tractor. It should be mentioned that the above-mentioned studies were devoted to single-row TRBs. For double-row TRBs, Nelias et al.[22] developed a model to analyze the bearing fatigue life by considering the effect from axial compression and tilting moment. Yang et al.[23] studied the fatigue life of a DTRB under combined loads and angular misalignment, but did not consider the effect of contact stress concentration on fatigue life. Lostado et al.[24] proposed a finite element model to determine the contact stress in a DTRB, which could be useful for the fatigue life analysis. Recently, Zheng et al.[25,26] proposed a five-DOF model of a DTRB by taking the thrust frictional force on the roller into account. The internal load and contact pressure distribution of rollers were then analysed in relation to roller profile and angular misalignment.[25]

From the above literature review, it can be seen that the previous studies were mainly focussed on the fatigue life prediction for bearings under constant load and continuous speed. For the DTRBs used in floating direct-drive wind turbines, they are exposed to significant oscillating load and speed, which are induced by the winds, waves and buoyant forces from the floating platform. Aiming for the DTRBs in direct-drive wind turbines, this paper will investigate the fatigue life of a DTRB by considering oscillating external load and varying rotating speeds. To this end, a five-DOF model for DTRB developed earlier by the authors[25] is adopted first for finding the contact load and misalignment angle between rollers and raceways. From that, the contact pressure distribution along the rollers can be determined using a three-dimensional (3D) elastic contact model. To take the oscillating load and speed into account, a modification factor for the basic rating life is presented for fatigue analysis. The effects of operating conditions, angular misalignment, and internal clearance on the fatigue life of DTRB are then investigated using numerical simulations.

This paper is organized into five sections. The following section briefly discusses the contact loads and pressure distributions of rollers in DTRB. Fatigue life calculation is carried out in a later section. Further, numerical results and discussions are given. Concluding remarks are made in the last section.

## Contact loads and contact pressures of rollers in DTRB

Fatigue life estimation requires the calculation of the internal contact loads and contact pressures on rollers. In this section, the contact loads of rollers will be determined by solving the relevant equilibrium equations of DTRB using a quasi-static model.

### *Quasi-static model of DTRB*

The detailed description of the DTRB model was presented in literature[25] without considering the bearing clearances. This section briefly develops the bearing model particularly with the inclusion of radial and axial clearances. Figure 2 shows the cross-section of a DTRB with some basic geometric parameters and clearances. The inner ring of DTRB is subjected to a load vector $\{F\}$, which causes a displacement vector $\{\delta\}$ (Figure 3(a) and (b))

$$\{F\}^T = \left\{F_x, F_y, F_z, M_x, M_y\right\} \tag{1}$$

$$\{\delta\}^T = \left\{\delta_x, \delta_y, \delta_z, \gamma_x, \gamma_y\right\} \tag{2}$$

where the superscript $T$ represents the transpose of a vector.

Figure 3(c) shows the local coordinate system $(r, \psi, z)$ located at the inner ring cross-section at a particular roller of azimuth angle $\psi$. A cross-section view of the inner ring under the local coordinate system is shown in Figure 4(a). The displacement vector of the inner ring cross-section, displacement vector of roller, and the inner ring contact load vector are expressed respectively as

$$\{u_M\}^T = \left\{u_{M,r}, u_{M,z}, \theta_M\right\} = [R_M]\{\delta\} \tag{3}$$

$$\{v_M\}^T = \left\{v_{M,r}, v_{M,z}, \phi_M\right\} \tag{4}$$

$$\left\{Q_M\right\}^T = \left\{Q_{M,r}, Q_{M,z}, T_M\right\} \tag{5}$$

where the subscript $M$ is the row index, $M=1$ represents the right row, and $M=2$ the left row. Given the fact that the axis of roller is inclined by an angle $\alpha_m$, it would be more convenient to use an inclined coordinate system as shown in Figure 4(b). Then the corresponding displacement and load vectors shown in equations (3) to (5) become

$$\left\{u_{M,\alpha_m}\right\}^T = \left\{u_{M,\xi}, u_{M,\zeta}, \theta_M\right\} = [K_M]\{u_M\} \tag{6}$$

$$\left\{v_{M,\alpha_m}\right\}^T = \left\{v_{M,\xi}, v_{M,\zeta}, \phi_M\right\} = [K_M]\{v_M\} \tag{7}$$

$$\left\{Q_{M,\alpha_m}\right\}^T = \left\{Q_{M,\xi}, Q_{M,\zeta}, T_M\right\} = [K_M]\left\{Q_M\right\} \tag{8}$$

where $[R_M]$ and $[K_M]$ denote the transformation matrices.[25] The displacements of roller and inner race can lead to the contact compressions of rollers

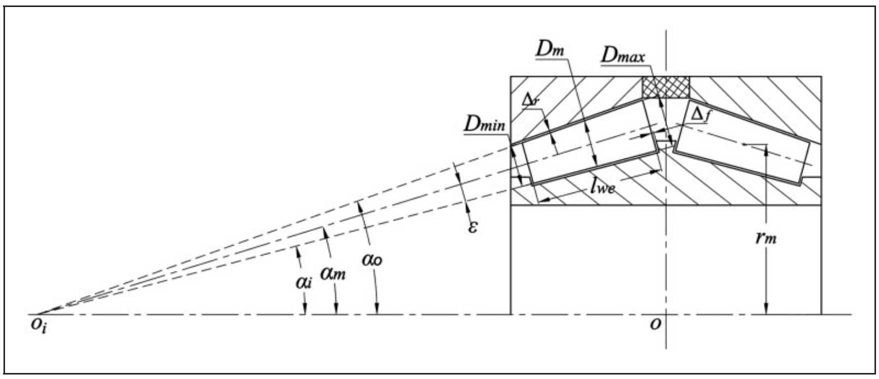


**Figure 2.** Basic geometric parameters of a DTRB.

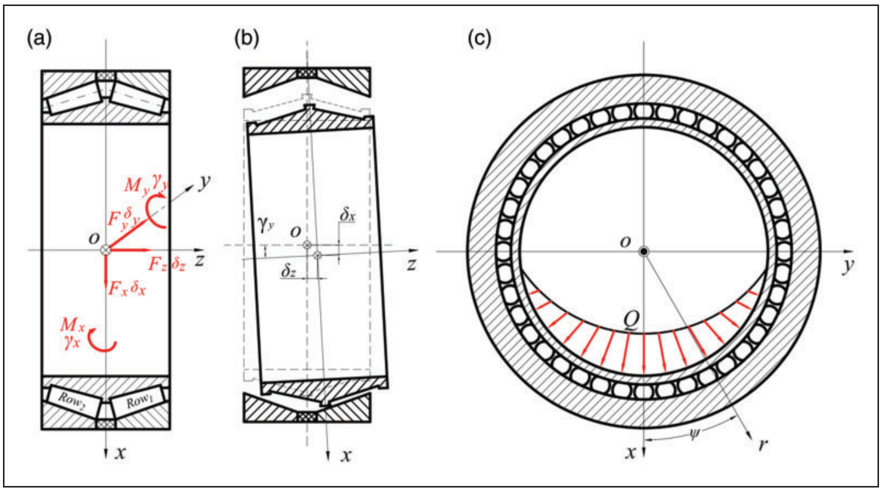


**Figure 3.** The five DOF model of DTRB: (a) the global coordinate system; (b) the angular misalignment of the inner ring; (c) the load zone and the local coordinate system at a particular roller.

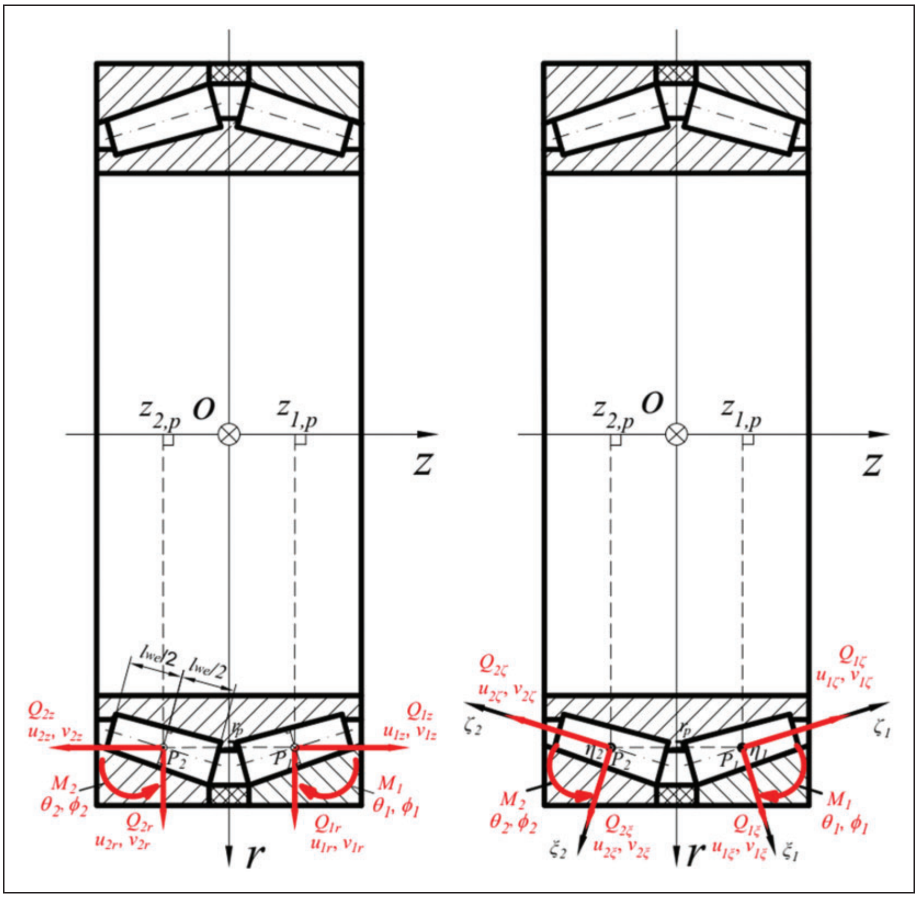


**Figure 4.** Loading and displacement representation of a local roller under: (a) the cylindrical coordinate system; (b) the inclined coordinate system.

which are calculated by

$$\delta_{M,i,k} = \left[\left(u_{M,\xi} - v_{M,\xi}\right)\cos\varepsilon - \left(u_{M,\zeta} - v_{M,\zeta}\right)\sin\varepsilon\right] + (\theta_M - \phi_M)l_k - h_k - \Delta_r \tag{9}$$

$$\delta_{M,o,k} = \left(v_{M,\xi}\cos\varepsilon + v_{M,\zeta}\sin\varepsilon\right) + \phi_M l_k - h_k - \Delta_r \tag{10}$$

$$\delta_{M,f} = [(u_{M,\xi} - v_{M,\xi})\sin\mu_0 - (u_{M,\zeta} - v_{M,\zeta})\cos\mu_0] + [\cos\mu_0 - \cos(\mu_0 + \theta_M - \phi_M)]l_f - \Delta_f \tag{11}$$

where $\delta_{M,N,k}$ and $\delta_{M,f}$ are the contact compressions between roller and raceway at the $k$th slice, and between roller-end and flange, respectively. The subscript $N$ indicates the inner and outer rings, $N = i, o$. $h_k$ is the drop of contact caused by profiled roller.[9] $\mu_0$ is the initial angle between flange contact line and the axis of roller, which is denoted by $_M$ after the external load is applied. $\Delta_r$ and $\Delta_f$ impose the effects of radial clearance ($\Delta_x$) and axial clearance ($\Delta_z$) (as shown in Figure 2), which are given by

$$\Delta_r = \frac{\Delta_x}{2}\cos(\alpha_m + \varepsilon) + \frac{\Delta_z}{2}\sin(\alpha_m + \varepsilon) \tag{11}$$

$$\Delta_f = \frac{\Delta_x}{2}\sin(\alpha_m - \mu_M) + \frac{\Delta_z}{2}\cos(\alpha_m - \mu_M) \tag{12}$$

Figure 5 demonstrates the contact force ($Q_{M,N}$) and moment ($M_{M,N}$) at the roller and raceway contact, which can be determined using the slicing method.[19] The contact force between the sphere-end of the roller and the flat flange of the inner ring ($Q_{M,f}$) can be obtained using Hertzian contact theory as

$$Q_{M,N} = \sum_{k=1}^{n_s} q_{M,N,k} = \sum_{k=1}^{n_s} c\delta_{M,N,k}^{10/9}\Delta l_k; \quad \left(\delta_{M,N,k} > 0\right) \tag{13}$$

$$M_{M,N} = \sum_{k=1}^{n_s} q_{M,N,k} l_k = \sum_{k=1}^{n_s} c\delta_{M,N,k}^{10/9}\Delta l_k l_k; \quad \left(\delta_{M,N,k} > 0\right) \tag{14}$$

$$Q_{M,f} = c_f \delta_{M,f}^{3/2}; \quad \left(\delta_{M,f} > 0\right) \tag{15}$$

where parameters $c$ and $c_f$ represent the contact constants.[27] $\Delta l_k$ and $l_k$ denote the length of a contact slice and the axial position of the $k$th slice, respectively. Under the above loads, the equilibrium equation for a roller can be established as given by equation (16), which is solved to find the roller displacement vector $\{v_M\}$ and the inner ring contact load $\{Q_M\}$

$$\left\{ \begin{array}{l} \left\{ \begin{array}{l} \left(Q_{M,i} - Q_{M,o}\right)\cos\varepsilon + Q_{M,f}\sin\mu_0 \\ +\left(F_{t,M,i} - F_{t,M,o}\right)\sin\varepsilon \end{array} \right\} \\ \left\{ \begin{array}{l} \left(-Q_{M,i} - Q_{M,o}\right)\sin\varepsilon + Q_{M,f}\cos\mu_0 \\ +\left(F_{t,Mi} + F_{t,M,o}\right)\cos\varepsilon \end{array} \right\} \\ M_{M,i} - M_{M,o} + Q_{M,f} l_f \sin\mu_M + \frac{F_{t,M,i} - F_{t,M,o}}{2\cos\varepsilon} \end{array} \right\} = \left\{ \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right\} \tag{16}$$

where $F_{t,M,N}$ indicates the thrust frictional force on the roller as shown in Figure 5.[25] The roller contact loads can be found by solving equilibrium equations for all rollers. Then, the displacement $\{\delta\}$ of the inner ring can be obtained by solving the global equilibrium equation of the inner ring

$$\{F\} + \sum_{M=1}^{2}\sum_{j=1}^{Z}[R_M]^T\{Q_M\} = \{0\} \tag{17}$$

The iterative Newton-Raphson method is used to solve the solution to the equilibrium equations, which are nonlinear. Figure 6(a) shows the computational procedure to solve the equilibrium equations of the DTRB. The process consists of two main iteration loops for solving the local equilibrium equations of rollers and of the inner ring. At the end of each iteration loop, the increments of roller displacement $\left(\Delta v_{M,\alpha_m}\right)$ and of the inner ring displacement ($\Delta\delta$) determined by the Newton–Raphson algorithm are evaluated against the pre-designated tolerances for the convergence check. The tolerances for roller ($\varepsilon_v$) and inner race ($\varepsilon_\delta$) displacements are all set to be $10^{-5}$ (mm, rad). The stopping conditions of the loops are that the increments must be smaller than the designated tolerances. After the convergence is realized, a new loop is repeated if the stopping conditions are not satisfied. Figure 6(b) illustrates an example of the convergence of the bearing equilibrium using the iterative Newton–Raphson method in Figure 6(a). The DTRB

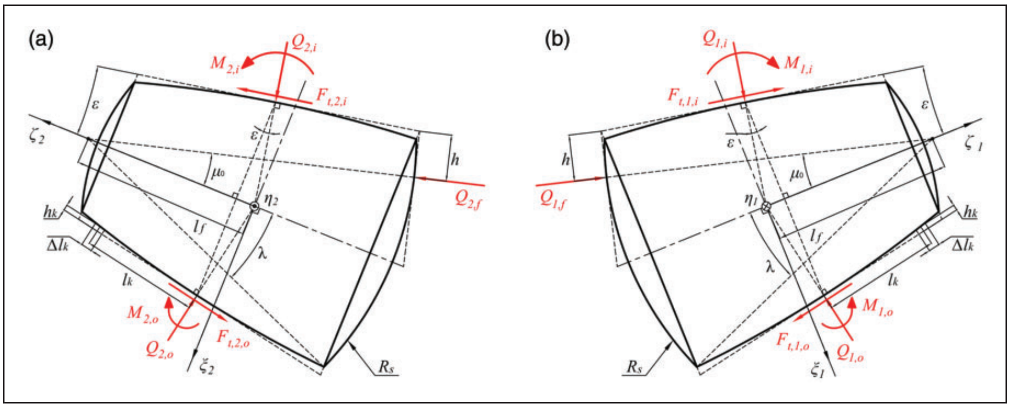


**Figure 5.** Free-body diagrams of rollers: (a) the left row (Row 2); (b) the right row (Row 1).

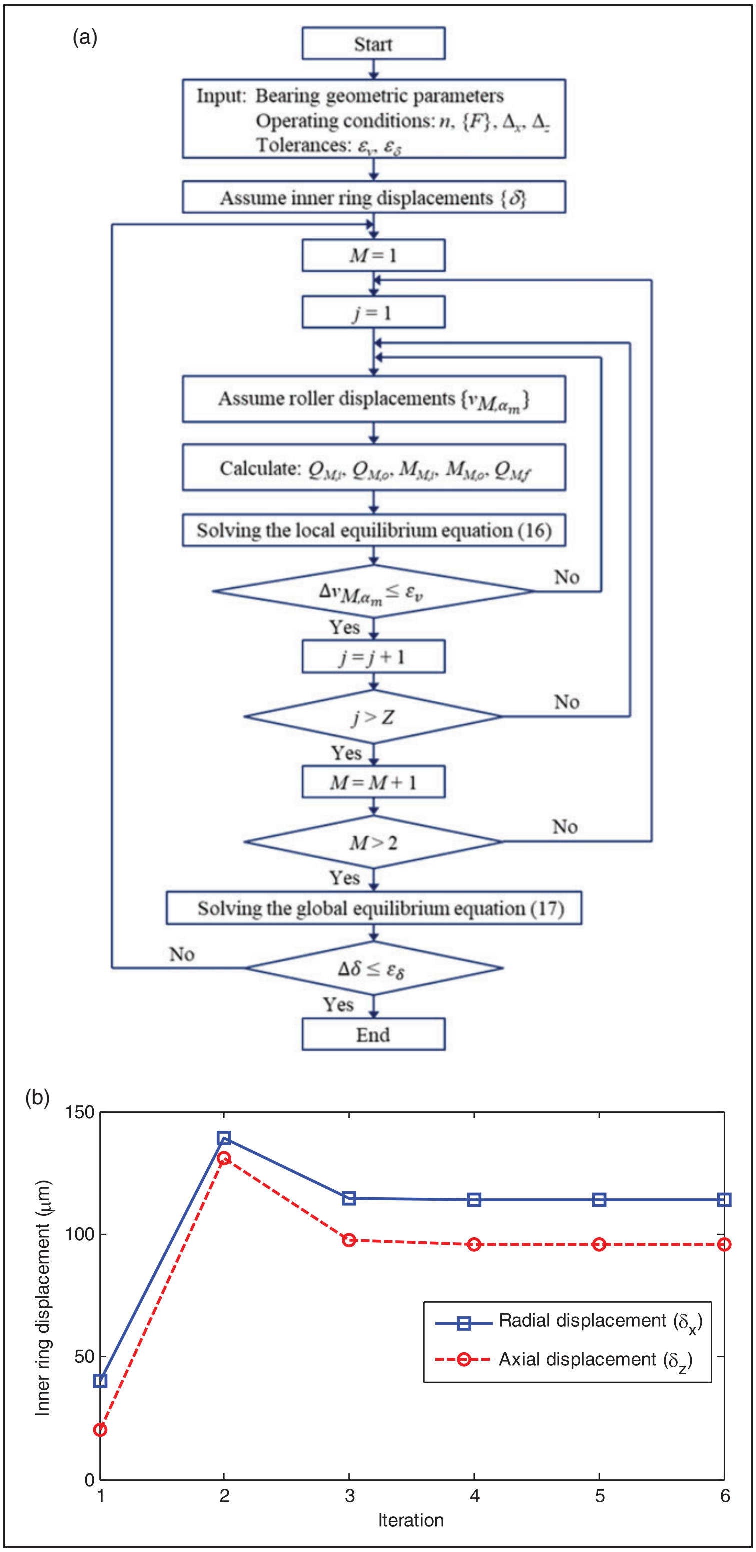


**Figure 6.** Computation process for bearing equilibrium and the convergence of the inner-ring displacements: (a) computation process for bearing equilibrium; (b) convergence of the inner ring displacements under the radial load $F_x = 3500$ kN and the axial load $F_z = 350$ kN.

is subjected to the combined radial and thrust loads of $F_x = 3500$ kN and $F_z = 350$ kN, and zero clearances. From Figure 6(b), it can be observed that the radial and axial displacements of the inner ring quickly converge to the steady-state values after six iterations.

### *Roller contact pressure distributions*

In order to determine the fatigue life of DTRB, it is necessary to calculate the contact pressure distribution in all the rollers. The 3D elastic contact approach is used to calculate contact pressure. To this end, the input is the roller contact force $Q$ and the relative misalignment angle between roller and raceway $\beta$, which are found by solving the quasi-static bearing model given in the previous section. Figure 7 presents the computational procedure of contact pressure distribution. Figure 8 shows an example of input $Q$ and $\beta$ calculated for the DTRB under the radial force $F_x = 3500\,\text{kN}$, the thrust force $F_x = 350\,\text{kN}$ and zero radial and axial clearances. First, the real contact area is assumed to be a rectangular area, which is then divided into $n_c$ small rectangular cells. Next, the initial gap of two contacting bodies $z_1$ and $z_2$ and the influence coefficients $f_{ij}$ are calculated. Subsequently, the contact pressures of all the $n_c$ rectangular cells and the approaching distance between roller and raceway are determined by solving a set of $n_c + 1$ linear equations. Among them, $n_c$ equations describe the dependence of the elastic deflection and distributed pressure over $n_c$ cells. The remaining equation reflects the relationship between the contact pressure and the normal contact load. Details of the equations are given in literature.[25] After obtaining the contact pressures in all the cells, the number of cells is re-calculated after eliminating the cells with negative pressure. Then, the iterative computational procedure is repeated until the contact pressures in all rectangular cells $p_j$ are positive.

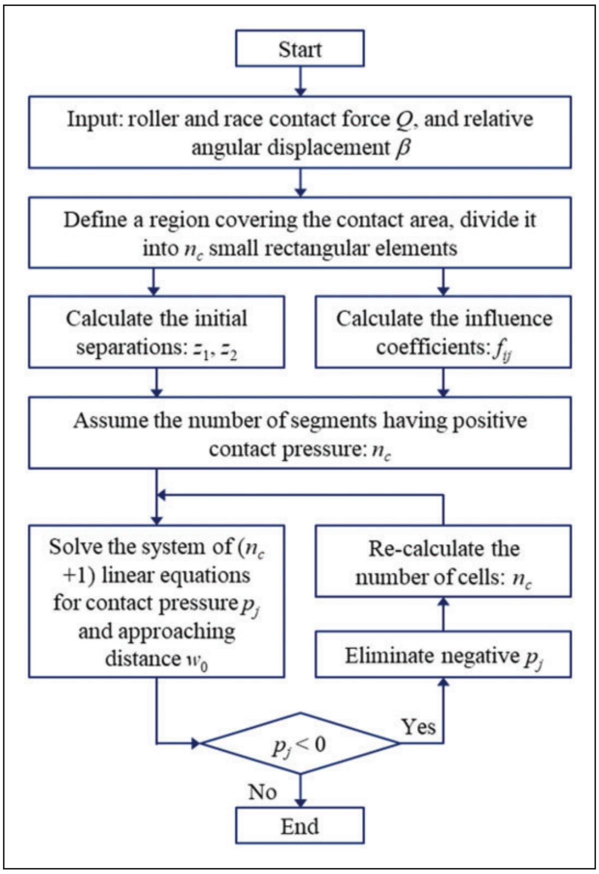


**Figure 7.** Computational procedure of contact pressure distribution.

Figure 9 shows the calculated contact pressure in the most highly loaded rollers shown in Figure 8 (the rollers at azimuth angle $\psi = 0°$). From the 3D contact pressure distribution shown in Figure 9(a) and (b), the corresponding maximum contact pressure profiles are found as shown in Figure 9(c), which are further used for fatigue life estimation. Due to the effect of axial load, the contact pressure of rollers located in Row 1 (the right row) is higher than that of Row 2 (the left row). The contact pressure of a roller with the inner and outer raceways are not equal, although their contact forces are almost identical (Figure 8). This is due to the difference in the contact geometry between roller and inner or outer raceway.

## Fatigue life calculation

In this section, the fatigue life of DTRB is analysed in terms of the modified basic reference rating life. For the case of invariant load and speed, the basic reference rating life ($L_{10\text{r}}$) can be calculated based on the ISO standard.[9] In order to determine the fatigue life under the oscillating load or speed, the common approach is to determine an oscillation factor $a_{osc}$ first. Then the modified basic reference rating life can be calculated by

$$L_{10rm} = a_{osc} L_{10r} \tag{18}$$

### *Basic reference rating life $L_{10r}$*

The basic reference rating life of a DTRB can be determined by[9]

$$L_{10r} = \left\{ \sum_{k=1}^{n_s} \left[ \left( \frac{q_{c,1,i}}{q_{ke,1,i}} \right)^{-9/2} + \left( \frac{q_{c,1,o}}{q_{ke,1,o}} \right)^{-9/2} + \left( \frac{q_{c,2,i}}{q_{ke,2,i}} \right)^{-9/2} + \left( \frac{q_{c,2,o}}{q_{ke,2,o}} \right)^{-9/2} \right] \right\}^{-8/9} \tag{19}$$

where $q_{c,M,N}$ is the basic dynamic load rating of a slice, which depends on the dimensions of the bearing. And $q_{ke,M,N}$ is the dynamic equivalent load of the $k$th slice, which is determined by the loading conditions of the bearing. $q_{c,M,N}$ and $q_{ke,M,N}$ are calculated by

$$q_{c,M,N} = q_{c,M,N} \left( \frac{1}{n_s} \right)^{7/9}, \quad (M = 1, 2 \text{ and } N = i, o) \tag{20}$$

$$q_{ke,M,N} = \left\{ \frac{1}{Z} \sum_{j=1}^{Z} \left[ \left( \frac{p_{\max\_N,j,k}}{271} \right)^2 D_m (1 \mp \gamma) \frac{l_{we}}{n_s} \right]^4 \right\}^{1/4}, \quad (M = 1, 2 \text{ and } N = i, o) \tag{21}$$

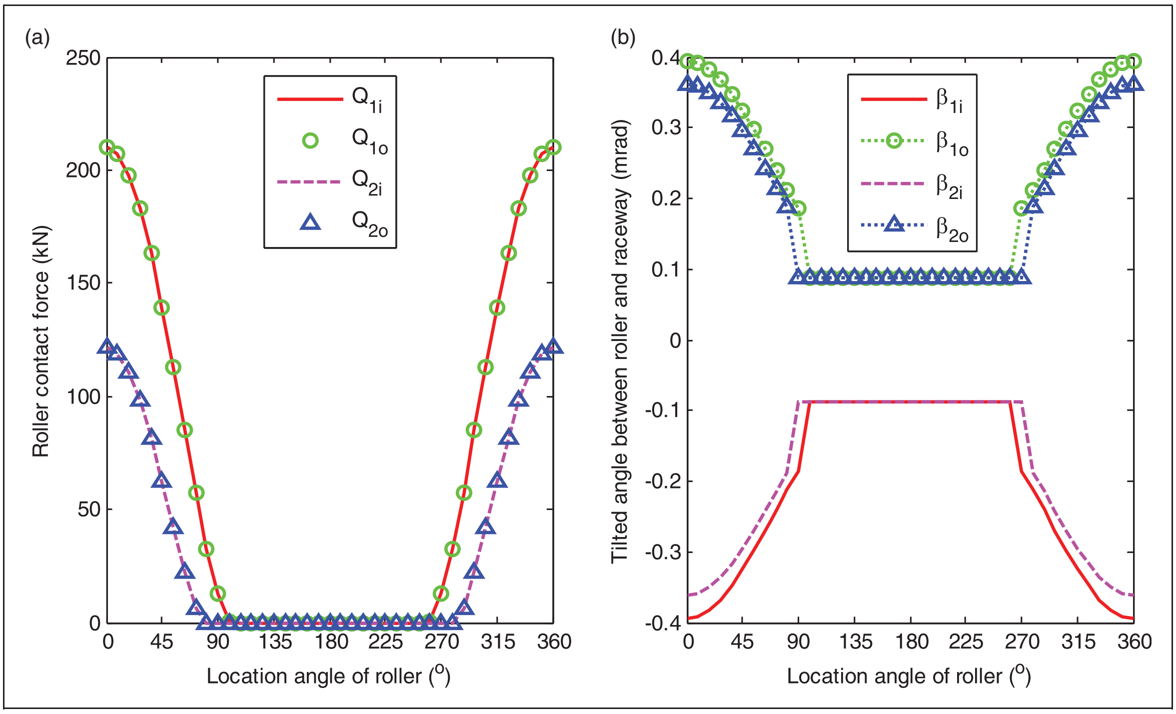


**Figure 8.** Contact force distribution among rollers and misalignment angle between rollers and raceways under the radial load $F_x = 3500$ kN and the thrust load $F_z = 350$ kN.

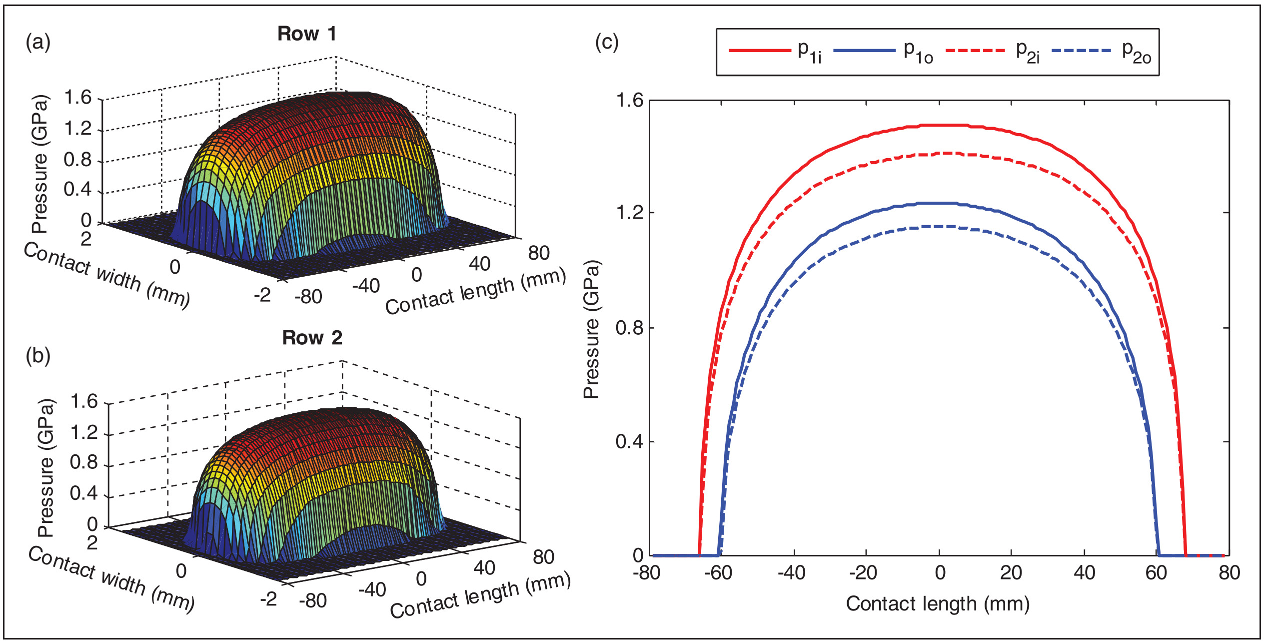


**Figure 9.** Pressure distribution in the most highly loaded roller under the radial load $F_x = 3500$ kN and the thrust load $F_z = 350$ kN: (a) 3D contact pressure of the roller with the inner race in Row 1; (b) 3D contact pressure of the roller with the inner race in Row 2; (c) the maximum contact pressure profiles of rollers.

where $p_{\max\_N,j,k}$ is the peak contact pressure of the $k$th slice of a roller $j$, which can be obtained from roller contact pressure calculation in an earlier section. $Q_{c,M,N}$ indicates the basic dynamic load rating of a roller which is expressed as[9]

$$Q_{c,M,N} = \frac{1}{\lambda \upsilon} \frac{C_{r,N}}{0.378 Z r^{7/9} \cos \alpha_N}$$

$$\times\left\{1+\left[1.038\left(\frac{1-\gamma}{1+\gamma}\right)^{143/108}\right]^{\mp 9/2}\right\}^{2/9},$$
$$(M=1,2\,\text{and}\,N=i,o) \tag{22}$$

In equations (21) and (22), the upper and lower signs are used for the inner and outer rings, respectively. $r$ represents the number of rows, $r=2$. The reduction factors $\lambda\upsilon=0.83$ and $\gamma=D_m\cos(\alpha_N)d_m$ are adopted from the literature.[9] Here, $D_m$ and $d_m$ are the mean roller diameter and pitch diameter of the bearing, respectively. The basic dynamic radial load rating of the DTRB is given by

$$C_{r,N}=b_m f_c(rl_{we}\cos\alpha_N)^{7/9}Z^{3/4}D_m^{29/27} \tag{23}$$

where the value of $f_c$ can be found by[9]

$$f_c=0.483B_1 0.377\lambda\upsilon\frac{\gamma^{2/9}(1-\gamma)^{29/27}}{(1+\gamma)^{1/4}}\times\left\{1+\left[1.04\left(\frac{1-\gamma}{1+\gamma}\right)^{143/108}\right]^{9/2}\right\}^{-9/2} \tag{24}$$

## *Oscillation factor*

Figure 10 shows the schematic of a bearing under oscillating motion. The amplitude of the oscillation is $\theta$. According to Harris et al.,[28] when the oscillation amplitude is higher than a certain critical angle ($\theta_{crit}$), the oscillation speed factor is given by

$$a_{osc\_n}=\frac{\pi}{2\theta} \tag{25}$$

If $\theta$ is smaller than $\theta_{crit}$, the oscillation speed factor is determined by

$$a_{osc\_n}=\left[\left(\frac{\pi}{2\theta}\right)^{2/9}Z^{0.0277}\right]^p \tag{26}$$

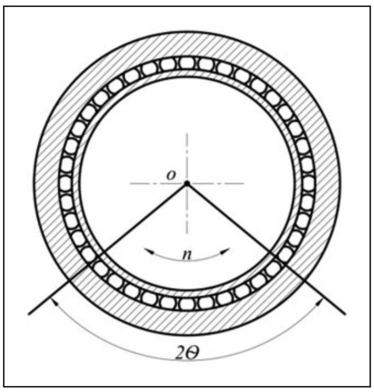


**Figure 10.** Oscillating motion of the bearing and the oscillating amplitude.

where $p=10/3$ is for roller bearing, and the critical angle is calculated by

$$\theta_{crit}=\frac{2\pi}{Z(1\pm\gamma)} \tag{27}$$

The oscillation factor in equation (26) is applicable for the case when the oscillation angle is larger than the so-called dither angle ($\theta_{dith}$). When the oscillation amplitude is smaller than $\theta_{dith}$, the fretting corrosion is likely to occur. This kind of failure is not a critical damage mechanism but it is unpredictable by using the ISO standard.[28] The value of $\theta_{dith}$ is given by

$$\theta_{dith}=\frac{2b}{d_m(1\pm\gamma)} \tag{28}$$

where $b$ is the width of the contact area between roller and raceway. In equations (27) and (28), the upper and lower signs are used for indicating the inner race and outer race, respectively.

Apart from the oscillating motion, the oscillating load may occur at the same time with the oscillation motion of the bearing. In determining the fatigue life with oscillating load, Harris and Kotzalas introduced a load modification factor $\psi_m$

$$\psi_m=\frac{\left\{\frac{\int_0^{2\pi}[P_s+P_d\sin(\varphi+\Delta\phi)]^p|\sin\varphi|d\varphi}{\int_0^{2\pi}|\sin\varphi|d\varphi}\right\}^{1/p}}{P_s} \tag{29}$$

where $P_s$ and $P_d$ are the steady value and the amplitude value of the oscillating load (see Figure 11). The relationship between $P_s$ and $P_d$ can be described via a load ratio $\beta$ as[27,29]

$$\beta=\frac{P_s}{P_s+P_d} \tag{30}$$

where the load ratio $\beta$ varies between 0 and 1, i.e. $0\leqslant\beta\leqslant 1$. When $\beta$ reaches to 1, the load becomes invariant and equals to $P_s$. $\Delta\phi$ is the phase shift angle, which demonstrates the phase difference between the oscillating load and speed. The oscillation

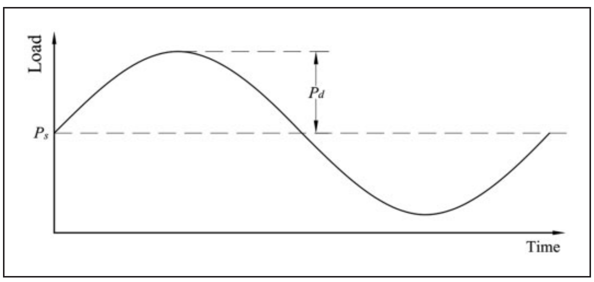


**Figure 11.** Oscillating load acting on the bearing.

load factor is calculated by

$$a_{osc\_p} = \psi_m^{-p} \tag{31}$$

The final modification factor can be obtained by combining $a_{osc\_n}$ and $a_{osc\_p}$ as

$$a_{osc} = a_{osc\_n} a_{osc\_p} \tag{32}$$

## Numerical results and discussion

Numerical simulations of the fatigue life are performed for a typical DTRB. Table 1 provides the geometry and material properties of the DTRB (LM287649D). First, the basic rating life with invariant load and speed is computed and the results are compared with an existing method to validate the proposed method. Then, the effects of the oscillating load and speed on the bearing fatigue life are investigated for the DTRB subjected to oscillating loads.

### *Basic reference rating life under invariant load and speed*

All the simulations are carried out under a constant rotational speed of $n = 12.1$ r/min. Figure 12 shows the calculated $L_{10r}$ as a function of the radial and axial loads. In this figure, the computed basic reference rating life ($L_{10r}$) with invariant load and speed is compared with that from Lundberg and Palmgren (LP) theory, in which $L_{10r}$ is calculated using[8]

$$L_{10r} = \left(\frac{C}{F}\right)^{10/3} \tag{33}$$

where $C$ is the basic dynamic load rating of the bearing and $F$ is the equivalent load.[8] As shown in Figure 12, the calculated $L_{10r}$ by the proposed method and LP theory is well matched.

It should be noted that equation (33) is valid only for a properly aligned bearing and no stress concentration caused by loading. In order to calculate $L_{10r}$ for general loading cases, such as with the angular misalignment, the method presented in the earlier section can be used. In practice, the angular misalignment frequently occurs, especially in many high load applications. Figure 13 shows the computed $L_{10r}$ with the varying misalignment angles from −4 to 0 mrad, in which the fatigue life is displayed in a logarithmic scale. It is observed that $L_{10r}$ is reduced with increasing the misalignment angle. The reduction rate is decreased as the radial or thrust load rapidly increases.

Figure 14 shows the predicted fatigue life of the aligned DTRB with the varying axial and radial clearances. The axial clearance ($\Delta_z$) is changed from −300 to 300$\mu$m and the radial clearance ($\Delta_x$) from 0 to 60$\mu m$. The clearances with negative values are equivalent to the rigid preload of bearing. The radial and axial loads of bearing are fixed at $F_x = 2500$ kN, and $F_z = 100$ kN, respectively. As shown in Figure 14, a certain amount of the axial preload (negative clearance) can increase the bearing fatigue life. The fatigue life is reduced when the clearance is too large. Increasing the radial clearance tends to move the peak of $L_{10r}$ to the left-hand side, which means reducing axial preload. Therefore, the radial and axial preloads should be well controlled to obtain a high fatigue life. Figure 14(b) shows the deterioration of the fatigue life with an increase of the misalignment angle. The variation trend of $L_{10\mathrm{r}}$ under the clearance effect is similar for the bearing with and without misalignment.

Figure 15 shows the maximum contact pressure distribution along the inner race of the aligned and misaligned bearings. It is seen that the misalignment angle changes the contact pressure distribution along the raceway, with higher value of localized pressure. Especially, the contact pressure around the location angle $\Psi = 0°$ experiences a significant increase under angular misalignment effect. Due to the increase in contact pressure, the bearing fatigue life with misalignment is reduced as shown in Figures 13(b) and 14(b).

**Table 1.** Basic parameters of the DTRB.[25]

| Parameter | Symbol, unit | Value |
|---|---|---|
| Bore diameter | $d$ | 938.21 mm |
| Outer diameter | $D$ | 1270 mm |
| Inner ring width | $B$ | 400.05 mm |
| Outer ring width | $C$ | 169.86 mm |
| Number of rollers per row | $Z$ | 40 |
| Inner race contact angle | $\alpha_i$ | 10.8° |
| Tapered roller semi-cone angle | $\epsilon$ | 0.85° |
| Outer race contact angle | $\alpha_o$ ] | 12.5° |
| Roller effective contact length | $L_{we}$ | 158.75 mm |
| Diameter of roller large-end | $D_{\max}$ | 83.44 mm |
| Diameter of roller small-end diameter | $D_{\min}$ | 78.49 mm |
| Maximum inner raceway contact diameter | $d_{\max}$ | 1055.15 mm |
| Minimum inner raceway contact diameter | $d_{\min}$ | 994.95 mm |
| Roller elastic modulus | $E_1$ | 310 GPa |
| Ring elastic modulus | $E_2$ | 200 GPa |
| Roller Poisson's ratio | $v_1$ | 0.26 |
| Ring Poisson's ratio | $v_2$ | 0.25 |

### *Effect of oscillating load and speed on bearing fatigue life*

The effect of the oscillating speed and load condition on the modified basic rating life ($L_{10\mathrm{rm}}$) is investigated

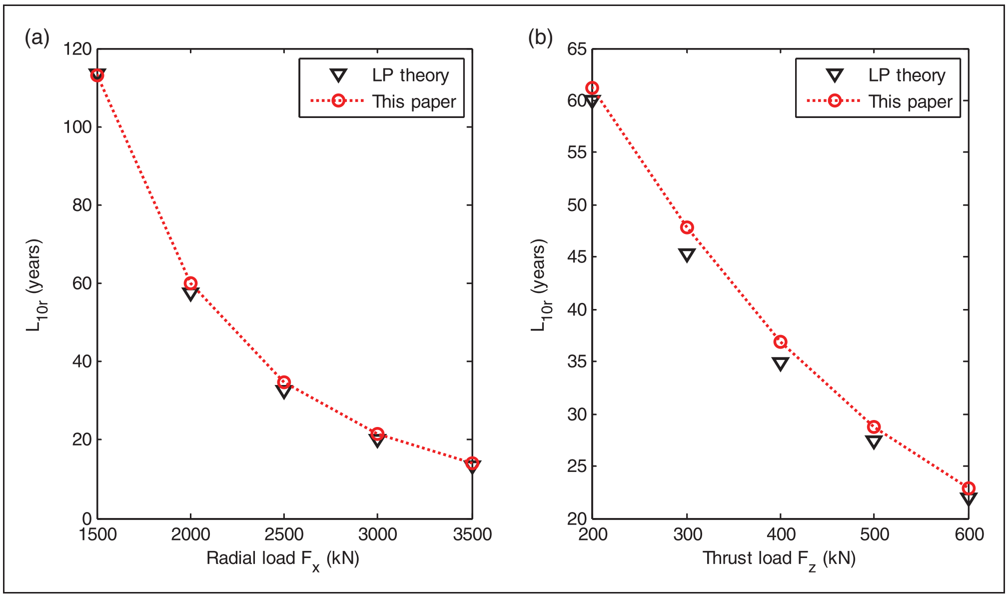


**Figure 12.** Comparison of $L_{10r}$ calculated by LP theory and the proposed method: (a) under the variation of radial load for the thrust load $F_z = 300$ kN; (b) under the variation of thrust load for the radial load $F_z = 2200$ kN.

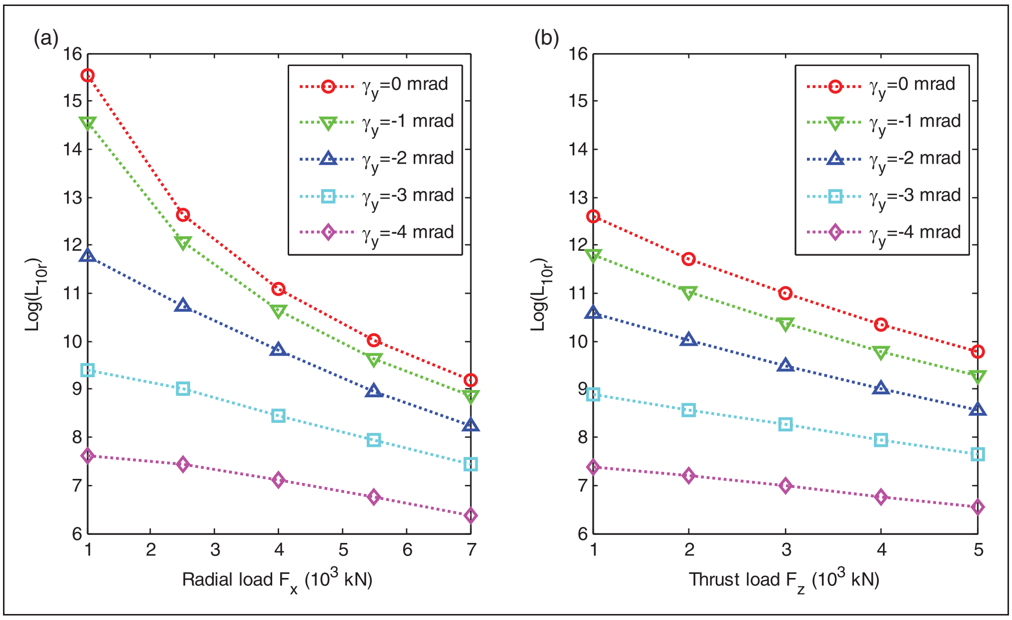


**Figure 13.** The variation of $L_{10r}$ with angular misalignments and loads: (a) with radial load for the thrust load $F_z = 100$ kN; (b) with thrust load for the radial load $F_x = 4000$ kN.

for the DTRB with the radial clearance $\Delta_x = 30\,\mu$m, and the axial clearance $\Delta_z = -150\,\mu$m. The effect of oscillating radial load of the bearing is first considered. The radial load $F_x$ is assumed to follow a periodic sinusoidal oscillation around a mean load $P_s$ as shown in Figure 11.[27] The thrust load of the DTRB is kept constant at $F_z = 100$ kN. The load ratio $\beta$ is selected from 0.5 to 1.0. From $P_s$ and $\beta$, the oscillating load amplitude $P_d$ can be calculated using equation (30).

Figure 16 shows the effects of oscillation speed and radial load on the fatigue life of DTRB. The corresponding fatigue life estimated for invariant radial load $F_x = P_s = 2500$ kN and speed n = 12.1 r/min is $L_{10r} = 35.58$ years for the aligned DTRB, and $L_{10r} = 20.54$ years for the misaligned DTRB. The $L_{10rm}$ with oscillating speed is higher than $L_{10r}$, as shown in Figure 16(a). $L_{10rm}$ reduces nonlinearly with the increase of oscillating amplitude $\theta$. Theoretically, when $\theta$ becomes very large, the motion of bearing is approximated to a continuous motion and therefore, the $L_{10rm}$ decreases to reach $L_{10r}$. Figure 16(b) shows that $L_{10rm}$ with oscillating load experiences an obvious reduction when

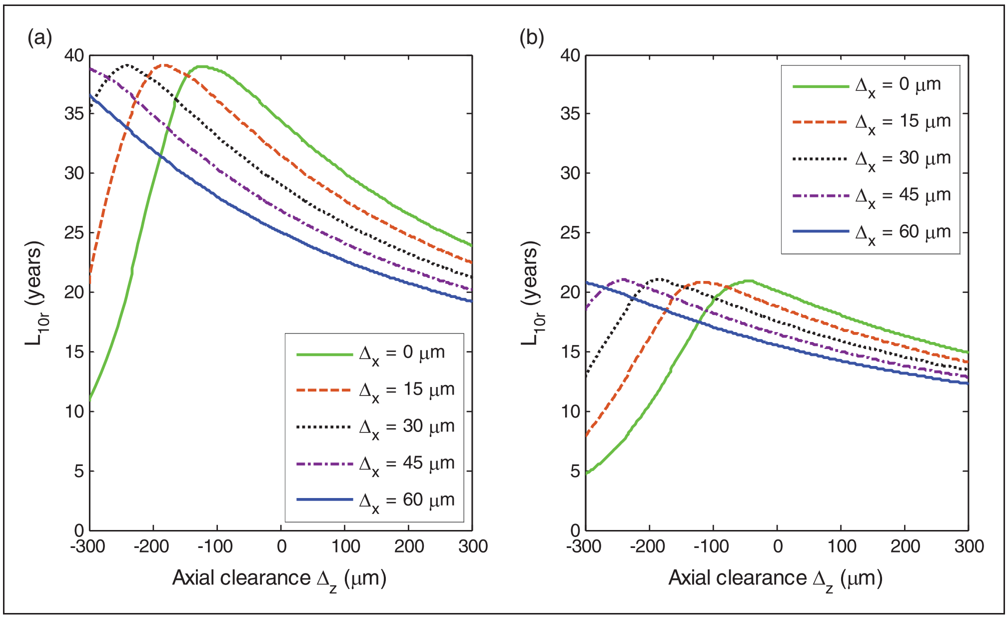


**Figure 14.** $L_{10r}$ with varying radial and axial clearances under the radial load $F_x = 2500$ kN and thrust load $F_z = 100$ kN for: (a) the aligned DTRB; (b) the misaligned DTRB of –1 mrad.

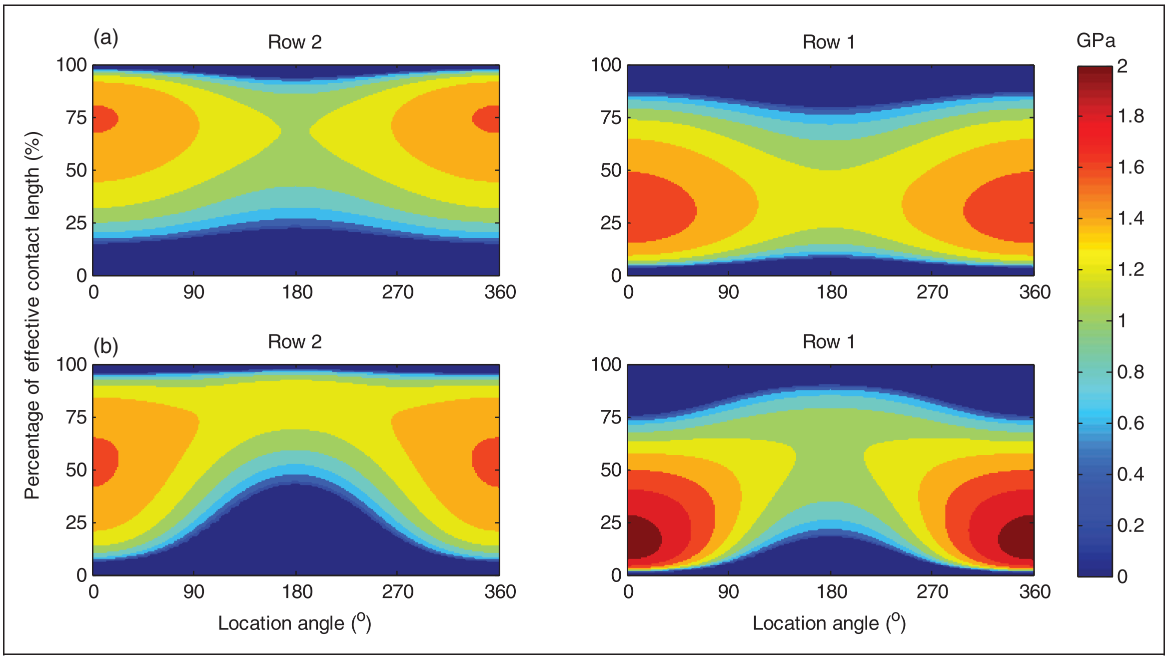


**Figure 15.** Maximum contact pressure along the inner raceway of DTRB under the radial load $F_x = 2500$ kN, the thrust load $F_z = 100$ kN, and the clearances $\Delta_x = 60\,\mu$m, $\Delta_z = 200\,\mu$m for: (a) the aligned bearing; (b) the misaligned bearing of –1 mrad.

compared to $L_{10r}$. The smaller the load ratio $\beta$, the shorter the fatigue life. This is because a small value of $\beta$ indicates a higher oscillating load amplitude $P_d$ as indicated in equation (30). The fatigue life with oscillating load gradually reaches to $L_{10r}$ as the load ratio $\beta$ approaches 1.0.

Figure 17(a) and (b) shows the effect of the oscillating speed under different loads on the fatigue life of the aligned and misaligned DTRBs, respectively. The misalignment angle is –1 mrad. As the oscillating load and speed are taken into account simultaneously, the effect of phase-shift angle $\Delta\phi$, which indicates the difference between phase angles of speed and load, is considered. A range of $\Delta\phi$ from 0° to 90° is selected. The steady state value of the oscillating load and the load ratio are $P_s = 2500$ kN and $\beta = 0.75$, respectively. Figure 17 shows that at $\Delta\phi = 0$, i.e. the load and speed are in-phase, the

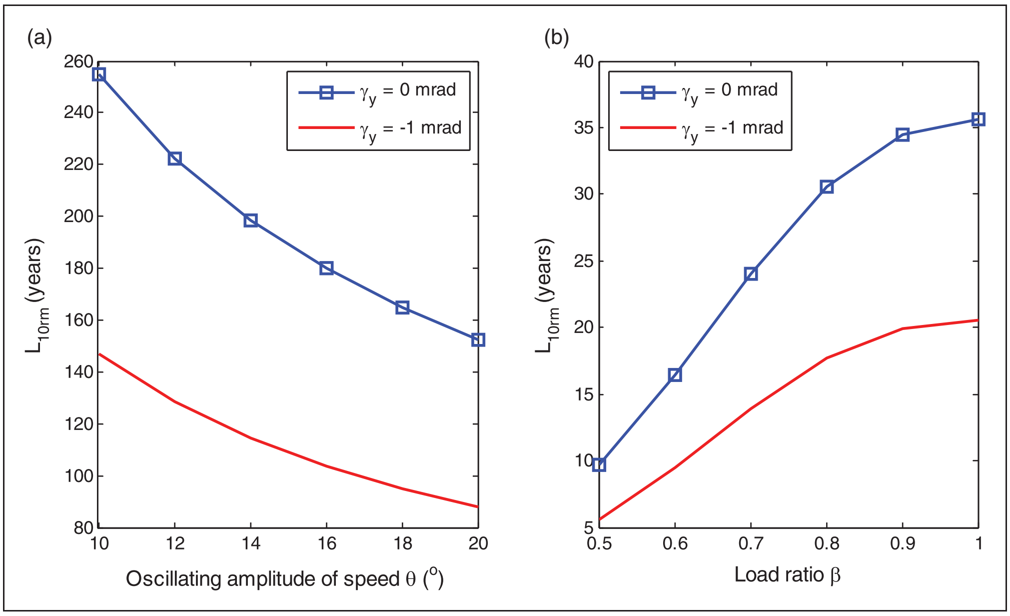


**Figure 16.** Effects of oscillating speed and load on the fatigue life of DTRB: (a) the oscillating speed; (b) the oscillating load.

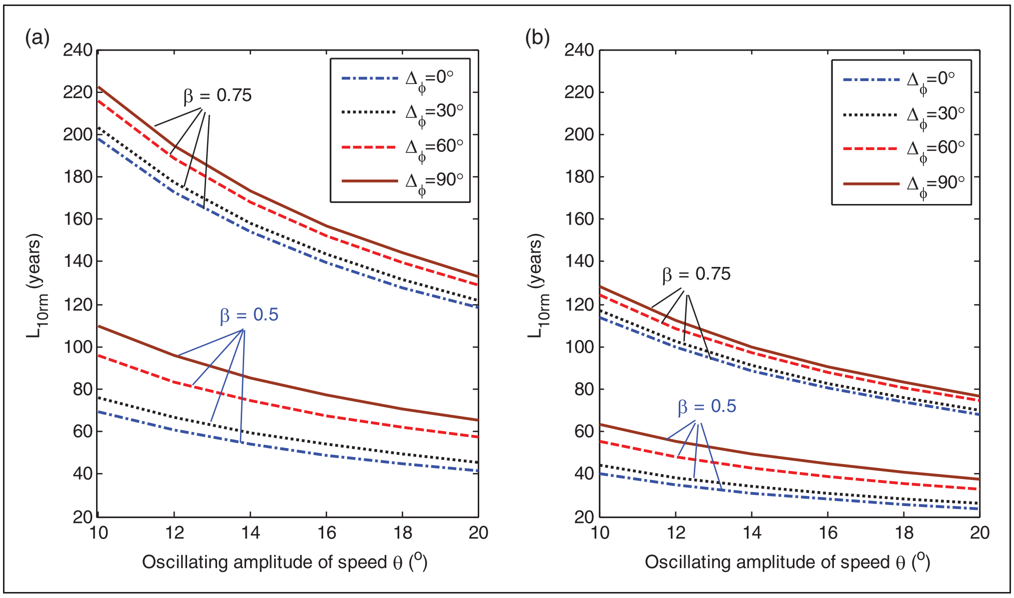


**Figure 17.** Effect of the oscillating speed on fatigue life with load ratio $\beta = 0.75$ and $\beta = 0.5$: (a) the aligned bearing; (b) the misaligned bearing of –1 mrad.

fatigue life of bearing is minimal. With increasing $\Delta\phi$, the fatigue life is also increased. The fatigue life in Figure 17(b) is lower than that in Figure 17(a) because of the inclusion of misalignment angle. Figure 17 also displays the effect of oscillating speed under different loads on the fatigue life for a smaller load ratio of $\beta = 0.5$. It is easy to notice that reducing the load ratio leads to the reduction of fatigue life compared to the case $\beta = 0.75$. This is caused by the increase in the oscillating load amplitude $P_d$. However, the variation trend of fatigue life with $\beta = 0.5$ is still similar to that with $\beta = 0.75$ when considering the effect of $\Delta\phi$.

## Concluding remarks

In this study, a mathematical model has been developed to calculate the fatigue life of a DTRB subjected to combined external radial and thrust loads, angular misalignment, and internal clearance. The oscillating load and speed was considered in the model to represent the practical operating situation of DTRB used in large modern floating direct-drive wind turbines. Numerical results obtained from the present proposed model showed a good agreement with the classical method under some preliminary operating conditions. The effects of external loads, angular misalignment, internal clearance, oscillating load, and speed on the bearing fatigue life were numerically investigated. The following concluding remarks can be made:

1. With the increase of the combined external load and misalignment angle, the fatigue life $L_{10r}$ is reduced. The reduction is significant as the radial or thrust load greatly increases.
2. The peak value of the predicted fatigue life $L_{10r}$ tends to move to the left-hand side on increasing the radial preload (equivalently reducing axial preload). Hence, the radial and axial preloads and clearance should be well controlled to obtain a high fatigue life.
3. Under oscillating speed and load conditions, reducing the oscillating amplitudes of speed and load is needed to improve the fatigue life of DTRB. Meanwhile, the fatigue life of bearing is increased on increasing the phase shift angle $\Delta\phi$.

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